

Evan Tan

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EDUCATION

National University of Singapore

Aug 2026 - Jul 2030

Double Degree: B.Sc. in Computer Science & B.B.A. in Business Administration (Finance)

- Extracurricular: NUS Fintech Society (Machine Learning Tech Lead), NUS Product Management (Curriculum Executive), NUS C-Innovators programme, Residential College 4 (RC4).
- Activities: NUS-SDS Datathon 2026, Tiktok Tech Jam2026, IDEATE 2026.

WORK EXPERIENCE

10x.ai, FlyRank.AI | Machine Learning Engineer Intern

Jun 2026 – Aug 2026

- Architecting ML-driven SEO optimization pipelines and predictive models to automate organic growth. Under child company FlyRank.AI. I participated in a 7-week machine learning capstone project by executing a rigorous data-framing framework; defined units of analysis, operational actions, and false-positive risk metrics to justify the model's business value.

RSAF, RAiD | Forward Deployed Engineer Intern (Mission Data)

Oct 2025 – Jun 2026

- Architected an agentic permit processing pipeline on Palantir Foundry using Claude Opus/Sonnet to extract 19 structured fields from unstructured PDFs, mapping data to a 91-permit ontology and reducing manual NOTAM generation from hours to minutes.
- Engineered a multi-modal geospatial system integrating PyMuPDF, HSV color detection, and GPT-5.1 Vision to parse DMS coordinates, constructing 174 validated polygons via turf.js for real-time operational dashboard visualization.
- Resolved critical data corruption cascades by implementing defensive, idempotent ontology edits and strict LLM guardrails, eliminating 100% of data pollution across 1,600+ automated extraction records.
- Diagnosed and remediated a platform-level GeoShape serialization bug, successfully re-routing image extraction pipelines to recover and persist 59 mission-critical geospatial polygon geometries in production.
- Executed comprehensive system health audits across action types and Automate monitors, deploying a bulk Spark-based remediation pipeline with non-deterministic LLM retry logic to achieve an 84% first-pass geospatial recovery rate.

RSAF, RAiD | Software Engineer Intern (Power Platform CoE)

Jan 2026 – Jun 2026

- Architected BoldFace, a full-stack, compliance-driven training and audit platform using React, Microsoft Dataverse, and Azure AD (MSAL); resolved critical cross-account data leakage via strict per-user keyed persistence.
- Spearheaded the Self-Regulated Training (SRT) ecosystem, building custom React/TypeScript PCF controls and automating Dataverse schema deployments via Node.js within a unified monorepo.
- Built a distributed majority-consensus approval mechanism and deterministic state machine for 100+ personnel, ensuring strict governance compliance in DLP-restricted environments.
- Designed a real-time AI decision-support system for ATC using Faster-Whisper and Qwen 2.5 (0.5B), enabling low-latency, CPU-optimized semantic parsing.
- Optimized operational workflows by engineering PySpark ETL pipelines for resource allocation and automating zero-touch flight routing via Power Automate.

PROJECTS

Instant Check Out Credit Engine | NUS FINTECH SOCIETY

Aug 2026 – Dec 2026

- System Architecture & Deployment: Engineered a production-ready, containerized dual-model API using FastAPI and Docker to serve real-time credit decisions for BNPL and microloans with sub-100ms latency.
- Multi-Modal ML Pipelines: Fused an Optuna-optimized XGBoost engine for structured financial data with a DistilBERT NLP pipeline that extracts dense semantic embeddings from unstructured transaction logs.
- Latency Optimization: Aggressively cached frequent token embeddings and applied heavy model distillation to the Transformer network, ensuring computationally intensive NLP tasks met strict e-commerce checkout latency requirements.
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Real-Time Navigational System For Visually Impaired | IDEATE 2026 Hardware Track

Jan 2026 – Feb 2026

- Concurrent System Architecture: Engineered a dual-core ESP32 processing hub utilizing FreeRTOS for strict task isolation, preventing system bottlenecks and guaranteeing high-availability execution for continuous 100 Hz sensor polling.
- Algorithmic Sensor Fusion: Developed a Complementary Filter to fuse raw MPU6050 IMU and TFmini Plus LiDAR telemetry via I2C/UART, executing real-time trigonometric projections ($\Delta h = \Delta d * \cos(\theta)$) to cancel gyroscope drift and calculate true vertical depth.
- Resource & Power Optimization: Optimized embedded power pipelines to a minimal 150 mA continuous draw, maximizing operational uptime to over 6.5 hours while architecting a highly scalable, low-cost production footprint.

SKILLS

- **Platforms & Architecture:** Palantir Foundry (AIP Logic, Workshop, Ontology Manager, Pipeline Builder), Microsoft Dataverse, Power Platform, Agentic Workflows, CI/CD, Application Lifecycle Management (ALM).
- **Languages:** Python, TypeScript, React, VBA, JavaScript, HTML/CSS, GO, SQL.
- **AI & Machine Learning:** LLM Integration (Claude Opus/Sonnet, GPT-5.1 Vision, Qwen), Faster-Whisper, Prompt Engineering, RAG, Layout-aware OCR, Scikit-learn, XGBoost.